

Adaptive MSCC Charging for Lithium-Ion Batteries with SoC-Dependent PSO–Fuzzy Optimization

Abstract—Rechargeable lithium-ion batteries have become the dominant energy storage solution in modern applications. However, conventional constant-current/constant-voltage (CC-CV) charging strategies face significant limitations in fast-charging applications, as the CV stage often prolongs the overall charging time and reduces energy efficiency. To overcome this issue, this paper proposes an adaptive multi-stage constant current (MSCC) charging strategy that optimizes the charging process based on the initial state of charge (SOC). The approach integrates particle swarm optimization (PSO) with fuzzy logic control to determine SOC-dependent stage parameters, thereby improving charging efficiency and reducing electrochemical stress. Simulation results demonstrate that the proposed method achieves a significant reduction in charging time—up to nearly 320 s compared with the conventional CC-CV method and 280 s compared with the traditional MSCC optimization designed only for 0% SOC. This improvement is obtained with only a minor trade-off in delivered capacity, i.e., approximately 30 mAh compared to CC-CV and 20 mAh compared to conventional MSCC. Furthermore, the results indicate that the initial charging current can be reduced while still maintaining optimized charging time, which helps alleviate current-induced stress on the battery and potentially extend its cycle life.

Index Terms—CC-CV, MSCC, SOC, Optimization, PSO.

I. INTRODUCTION

Rechargeable batteries play a crucial role in modern energy storage solutions for advanced technologies such as hand-held consumer electronics, aerospace systems, electric vehicles, and energy backup in renewable energy applications. Among the widely used rechargeable battery chemistries (lead-acid, nickel-based, and lithium-ion (Li-ion)), Li-ion batteries have become the dominant choice in modern electronic applications [1], [2], [3], owing to their advantageous characteristics, including higher environmental compatibility, superior energy and power density, elevated open-circuit voltage (OCV), low self-discharge rate, reduced maintenance requirements, and immunity to the memory effect [4]. These properties not only make Li-ion batteries a preferred candidate for advanced storage systems but also highlight the importance of developing efficient charging algorithms to maximize their performance, safety, and cycle life.

Fast charging has always been a critical consideration in the design of battery-powered devices. Consequently, it has become a major research focus, aiming to achieve reduced charging time while maintaining high efficiency. Beyond charging speed, several other factors must be addressed during the charging process, including charging efficiency, current stress, battery lifetime, and the accurate estimation and control of the state of charge (SOC). The conventional constant-current/constant-voltage (CC-CV) charging method, although widely adopted in practice, presents significant limitations in fast charging applications. Specifically, the CV stage typically occupies the majority of the charging duration, thereby

reducing overall energy efficiency and limiting the potential for accelerated charging [5]. These challenges underline the necessity of exploring advanced charging algorithms capable of balancing charging speed, efficiency, safety, and battery longevity.

To address this issue, research efforts have focused on the development of advanced charging techniques. One of the widely studied methods is the multi-stage constant current (MSCC) charging strategy, which aims to improve charging efficiency and reduce charging time. The benefits of this strategy include less chemical reaction stress and shorter charging duration. The MSCC method can also be optimized based on different transition criteria between charging stages, such as time-based transition, SOC-based transition, voltage threshold-based criteria, and cut-off voltage-based criteria. Among these, the cut-off voltage-based approach is the most commonly applied [6]. To further improve MSCC performance, intelligent control algorithms have been introduced, including fuzzy control, neural networks, grey-prediction control, and genetic approaches. These methods optimize the number of MSCC stages, charging time, and delivered capacity [5], [7], [8].

However, the aforementioned methods mainly focus on optimization under a single assumption, namely that the battery is charged from an initial state of charge (SOC) close to 0 %. These approaches are limited to determining an optimal set of MSCC parameters for charging from 0 % SOC, without considering the variability of charging performance at different initial SOC levels (e.g., 10 % to 90 %). As a result, when the charging process starts from different SOC values, the previously optimized stage parameters may no longer yield optimal performance.

To overcome this limitation, the proposed method aims to develop an adaptive MSCC charging strategy capable of adjusting the stage parameters according to the initial SOC level. By incorporating SOC-dependent optimization, the method seeks to improve charging efficiency, reduce charging time, and maintain battery health across a wide range of initial SOC conditions.

The rest of this paper is organized as follows. Section II introduces the multi-stage constant current (MSCC) charging method and reviews several battery models, where the Thevenin model is selected due to its advantages. Section III formulates the optimization problem based on the Thevenin model and presents the proposed approach, which combines particle swarm optimization (PSO) with fuzzy logic control. Section IV provides simulation results to validate the proposed method. Finally, Section V concludes the paper and outlines future research directions.

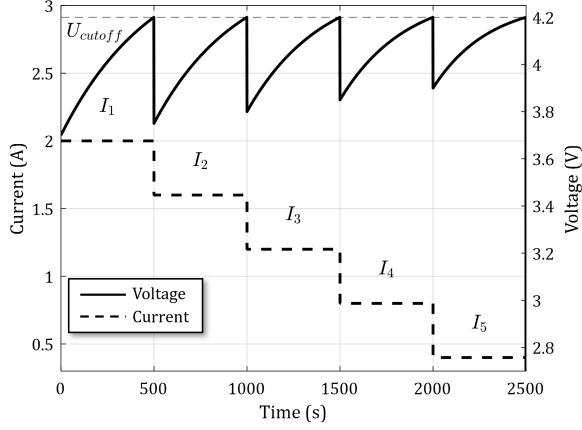


Fig. 1: Basic concept of the five-stage Multi-Stage Constant Current (MSCC) charging strategy.

II. MSCC ANALYSIS

A. Multi-Stage Constant Current (MSCC) Charging Strategy

The Multi-Stage Constant Current (MSCC) charging technique has been demonstrated in Fig. 1. In this method, the charging process is divided into multiple stages, where the charging current is gradually reduced as the battery voltage increases toward the cut-off threshold. In this work, a five-stage MSCC charging scheme is employed for lithium-ion batteries. Furthermore, the transition between stages is determined by predefined voltage thresholds, ensuring that each stage operates with an optimized current profile. That means once the terminal voltage reaches the threshold of that stage, the charging process proceeds to the next stage with a reduced current.

B. Equivalent Circuit Models

Equivalent circuit models (ECMs) are widely adopted due to their balance between accuracy and simplicity in describing Li-ion battery dynamics [9], [10]. Among them, three typical models are as follows:

1) Rint model: The simplest structure, consisting of an open-circuit voltage source U_{oc} and an ohmic resistance R_o . The terminal voltage can be expressed as:

$$U_L = U_{oc} - I_L R_o \quad (1)$$

Although easy to implement, the Rint model fails to capture transient polarization effects.

2) Thevenin model: An extension of the Rint model by adding a parallel RC network to represent polarization dynamics. Its electrical behavior is given by:

$$\begin{cases} \dot{U}_{Th} = -\frac{U_{Th}}{R_{Th}C_{Th}} + \frac{I_L}{C_{Th}} \\ U_L = U_{oc} - U_{Th} - I_L R_o \end{cases} \quad (2)$$

This model provides a balance between simplicity and accuracy, and is therefore widely adopted in battery management systems (BMS).

3) Dual Polarization (DP) model: To further improve accuracy, two RC branches are employed to separately represent

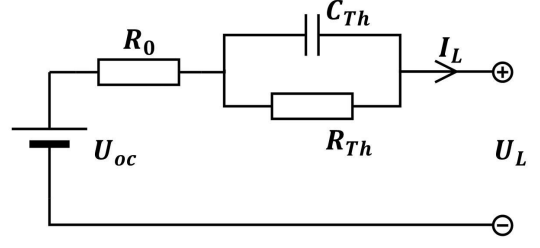


Fig. 2: Thevenin Equivalent Circuit Model

electrochemical polarization and concentration polarization. The model can be described as:

$$\begin{cases} \dot{U}_{pa} = -\frac{U_{pa}}{R_{pa}C_{pa}} + \frac{I_L}{C_{pa}} \\ \dot{U}_{pc} = -\frac{U_{pc}}{R_{pc}C_{pc}} + \frac{I_L}{C_{pc}} \\ U_L = U_{oc} - U_{pa} - U_{pc} - I_L R_o \end{cases} \quad (3)$$

Battery models with higher accuracy often come at the cost of increased complexity. Among the commonly used models—Rint, Thevenin, and Dual Polarization (DP)—the Thevenin model provides a reasonable trade-off, offering moderate complexity with sufficiently high accuracy [11]. Although its accuracy is slightly lower compared to the DP model, it remains reliable for practical applications [9], [10]. Therefore, the Thevenin model is selected in this work for calculation and optimization. As illustrated in Fig. 2, the model consists of the open-circuit voltage U_{oc} , the ohmic resistance R_o , and the polarization branch with resistance R_{Th} and capacitance C_{Th} [11]

III. FORMULATION AND OPTIMIZATION PROBLEM

A. Mathematical foundation

From Fig. 2, the continuous model from Thevenin can be discretized using the forward Euler method [11] and described in (4), (5):

$$U_{Th}(k+1) = U_{Th}(k) \left(1 - \frac{\Delta t}{R_{Th}C_{Th}}\right) + \frac{I_L(k)}{C_{Th}}, \quad (4)$$

$$U_L(k) = U_{oc}(k) - U_{Th}(k) - I_L(k)R_o(k), \quad (5)$$

where $U_{oc}(k)$, $R_o(k)$, $R_{Th}(k)$, and $C_{Th}(k)$ are updated at each step as seventh-order polynomial functions of SoC(k).

The charging capacity is updated iteratively as (6):

$$Q(k+1) = Q(k) + I_L(k) \cdot \Delta t, \quad (6)$$

and the state of charge (SoC) is calculated by (7):

$$\text{SoC}(k) = \frac{Q(k)}{Q_0}. \quad (7)$$

To calculate the charging time and capacity based on the Thevenin battery model, a discrete-time iterative computation loop is developed, as illustrated in Fig. 3. Although this discretization approach does not yield absolute accuracy, it simplifies the computational procedure and is well-suited for computer-based implementation.

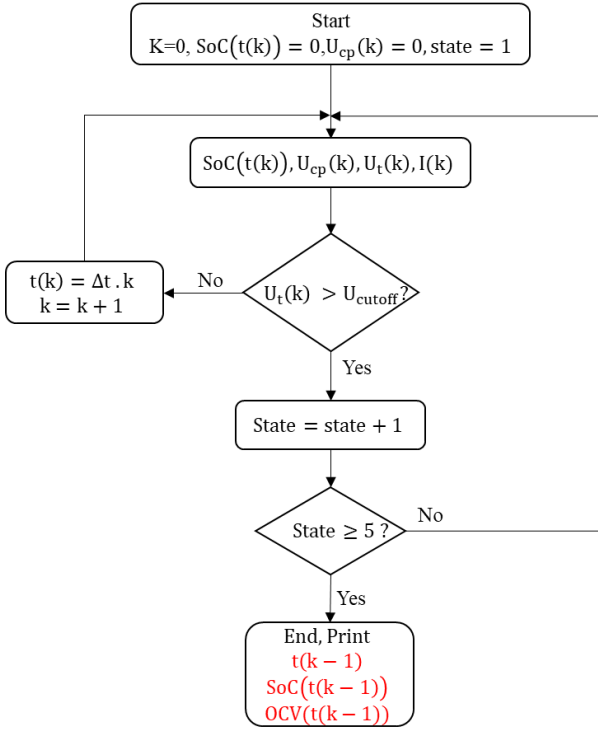


Fig. 3: Flowchart for calculating charging time using the MSCC method with the Thevenin model

B. Optimization Problem

According to the performance evaluation criteria of the MSCC charging method discussed earlier, and due to the limitations in time and research scope, this work focuses on optimizing two main criteria: *charging time* and *charged capacity*, with the aim of improving the overall MSCC efficiency. Therefore, the objective function can be formulated as (8):

$$\begin{aligned}
 &\textbf{Maximize: } F(T_{ch}(\vec{I}_i), C_{charge-norm}(\vec{I}_i)), \quad \vec{I}_i \in \mathbb{S} \\
 &\text{Where } \vec{I}_i = [I_1, I_2, I_3, I_4, I_5] \\
 &\textbf{Subject to: } 0 \text{ min} \leq T_{ch}(\vec{I}_i) \leq 90 \text{ min} \\
 &80\% \leq C_{charge-norm}(\vec{I}_i) \leq 100\% \\
 &I_i \geq I_j \quad \text{if } i < j, \quad i, j = 1, 2, \dots, 5
 \end{aligned} \tag{8}$$

In (8), $\mathbb{S}$ denotes the set of all feasible charging profiles. The objective is to maximize the cost-effectiveness function F , which incorporates two performance indices: charging time (T_{ch}) and normalized charged capacity ($C_{charge-norm}$, NCC). Here, C_{c-MSCC} refers to the charged capacity obtained using the MSCC method, while C_{c-CCCV} refers to the charged capacity obtained using the standard CCCV method. The normalized charged capacity is defined in (9):

$$NCC = \frac{C_{c-MSCC}}{C_{c-CCCV}} \tag{9}$$

Details of the construction of the objective function F are presented in the following section. In (8), all constraints related to the optimization problem are incorporated into

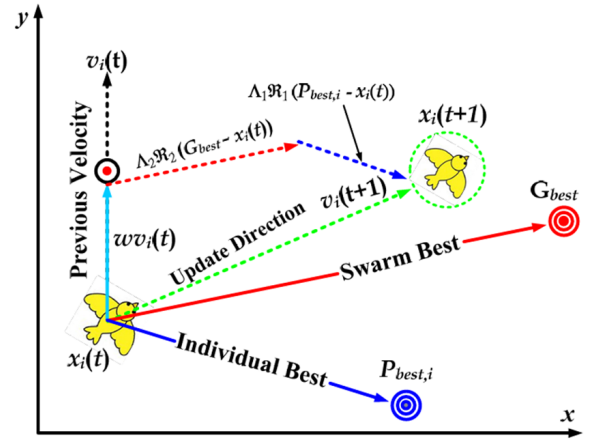


Fig. 4: Concept of particle update in PSO search

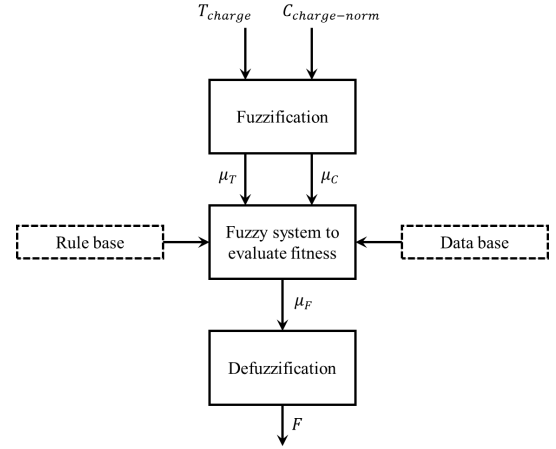


Fig. 5: FDFE structure for fitness evaluation.

the objective function in the form of penalty terms. The constraints on CT and NCC in this study are set within the ranges $[0, 90]$ minutes and $[80\%, 100\%]$, respectively, to meet general fast-charging requirements. Moreover, the charging current at a given stage must not exceed that of the preceding stage to reduce electrochemical reaction stress and improve the charged capacity.

IV. OPTIMIZATION ALGORITHMS AND PROPOSED METHOD

A. Particle Swarm Optimization

Particle Swarm Optimization (PSO) was first introduced by Eberhart and Kennedy in 1995. It is an effective method for solving global optimization problems involving single- or multi-objective functions with multiple local optima. The main advantage of PSO lies in its simplicity and efficiency, since only two model equations are required. Several variants have been proposed for specific applications; however, the most widely used form remains the global model (G_{best}), proposed by Eberhart and Shi. [4]

As shown in Fig. 4, the motion of each particle in an n -dimensional search space is determined by its velocity and position vectors, denoted as $\vec{x} = [x_1, x_2, \dots, x_n]$ and $\vec{v} = [v_1, v_2, \dots, v_n]$, respectively. During the search process, each

particle exchanges information with the swarm, combining individual cognition and global social interaction. The particle with the best solution shares its position (G_{best}) with others. Each particle then updates its trajectory based on its personal best position (P_{best}) and the global best position (G_{best}), according to (10) and (11):

$$v_i^n(k+1) = \underbrace{\omega v_i^n(k)}_{\text{inertia}} + \underbrace{\Lambda_1 \mathfrak{R}_1 (P_{best,i}^n - x_i^n(k))}_{\text{cognitive}} + \underbrace{\Lambda_2 \mathfrak{R}_2 (G_{best,i}^n - x_i^n(k))}_{\text{social}} \quad (10)$$

$$x_i^n(k+1) = x_i^n(k) + v_i^n(k+1) \quad (11)$$

where $x_i^n(k+1)$ and $x_i^n(k)$ denote the current and previous positions of particle i in the n -th dimension, and $v_i^n(k+1)$ and $v_i^n(k)$ denote the corresponding velocities. ω is the inertia weight, Λ_1 and Λ_2 are the cognitive and social learning coefficients, and $\mathfrak{R}_1$, $\mathfrak{R}_2$ are uniformly distributed random numbers in $[0, 1]$. G_{best}^n denotes the globally best position, while $P_{best,i}^n$ denotes the best position previously found by particle i .

B. Design a Fuzzy System to Evaluate Fitness

The fitness function in the PSO-based optimization is defined to evaluate the charging pattern performance. The optimal pattern should maximize the charge capacity while minimizing the charging time. The fitness function is given as:

$$F(T_{ch}, C_{charge-norm}) = f_\alpha(T_{ch}) \cdot T_{ch}(\bar{I}_i) + f_\beta(C_{charge-norm}) \cdot C_{charge-norm}(\bar{I}_i) \quad (12)$$

where $f_\alpha(\cdot)$ and $f_\beta(\cdot)$ denote the weighting factors of charging time (CT) and normalized charging capacity (NCC), respectively. Since CT and NCC are highly dependent on the charging current and exhibit a trade-off relationship, fixed weighting (e.g., 50%) may not provide optimal results. To address this, a fuzzy-based dynamic fitness evaluation (FDFE) is proposed, as illustrated in Fig. 5.

As shown in Fig. 5, CT and NCC are the inputs, and the output is the fitness value F . The Mamdani inference method with centroid-of-area (COA) defuzzification is applied.

The membership functions (MFs) of CT, NCC, and fitness are shown in Fig. 6. Symmetric triangular MFs with 50% overlap are used. The universes of discourse are $[0, 90]$ minutes for CT, $[80\%, 100\%]$ for NCC, and $[0, 1]$ for the normalized output. Each input variable is mapped into five linguistic terms, resulting in 25 rules in the fuzzy rule base. The complete rules are listed in Table I, where higher charge capacity and shorter charging time correspond to lower fitness.

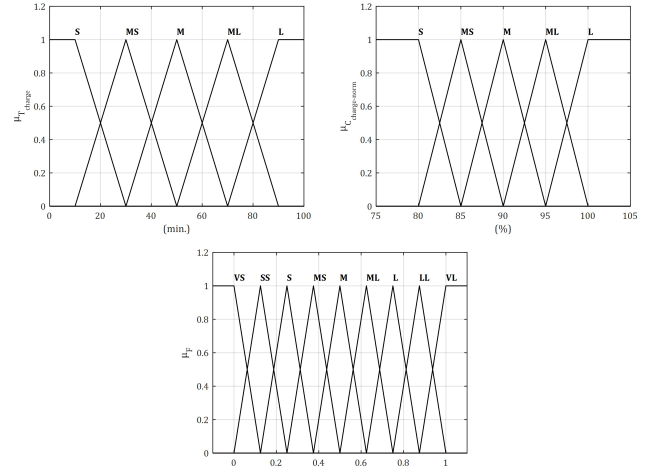


Fig. 6: Membership functions of the input/output variables.

TABLE I: Fuzzy inference rules for fitness evaluation

$C_{cha-norm} \backslash T_{ch}$	S	MS	M	ML	L
S	M Rule1	MS Rule2	S Rule3	SS Rule4	VS Rule5
MS	ML Rule6	M Rule7	MS Rule8	S Rule9	S Rule10
M	L Rule11	ML Rule12	M Rule13	MS Rule14	S Rule15
ML	LL Rule16	L Rule17	ML Rule18	M Rule19	MS Rule20
L	VL Rule21	LL Rule22	L Rule23	ML Rule24	M Rule25

Thus, the weighting allocation is adaptively determined by fuzzy rules instead of fixed coefficients, enabling effective incorporation of expert knowledge into the evaluation process.

C. Proposed Method

Figure 7 illustrates the flowchart of the fuzzy control search algorithm based on PSO applied in this paper. The main operational steps are described as follows:

Step 1: Measure OCV and then calculate initial SOC.

Step 2: Initialize the PSO parameters, including the inertia weight ω and the learning coefficients Λ_1 and Λ_2 . The initial charging samples for the first stage of each battery are generated randomly.

Step 3: Compute the charging time and the corresponding normalized charging capacity according to the iterative equations shown in the flowchart 6.

Step 4: The charging time and normalized charging capacity are fed into the fuzzy inference system to determine the fitness value of each particle in the swarm.

Step 5: Check the convergence condition: Has the maximum number of iterations been reached? If the condition is satisfied, the optimal solution will be obtained.

Step 6: If the convergence condition is not met, update the variables according to (10) and (11), then repeat Steps 3 to

5 until the convergence condition is satisfied and the optimal result is obtained.

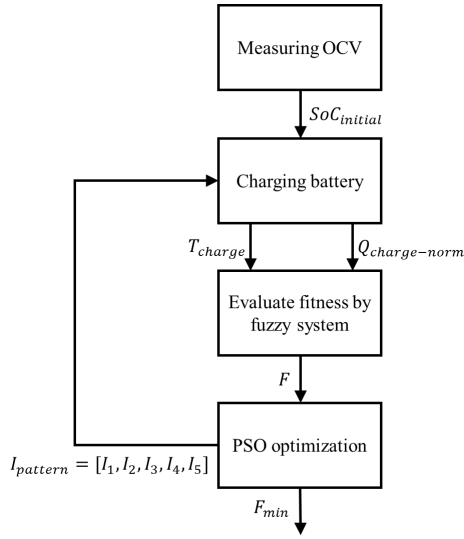


Fig. 7: Flowchart of the optimal charging current search process using PSO

V. SIMULATION RESULTS

Before performing the optimization, several parameters of the battery model are provided and summarized in Table III, based on the specifications of the Lithium-Ion LS LR1860LA cell manufactured by Lishen describe in Table II. In this work, the Thevenin battery model is employed in a 7th-order form and some parameters can be demonstrated from Table III, as follow (13), (14), (15), (16):

$$U_{oc}(k) = 2.896 + 9.563 SoC(k) - 57.92 SoC(k)^2 + 188.2 SoC(k)^3 - 340.7 SoC(k)^4 + 347.5 SoC(k)^5 - 184.2 SoC(k)^6 + 38.95 SoC(k)^7, \quad (13)$$

$$R_o(k) = 0.039 - 0.131 SoC(k) + 0.507 SoC(k)^2 - 1.129 SoC(k)^3 + 1.679 SoC(k)^4 - 1.764 SoC(k)^5 + 1.157 SoC(k)^6 - 0.331 SoC(k)^7, \quad (14)$$

$$R_{Th}(k) = 0.048 - 0.318 SoC(k) + 2.075 SoC(k)^2 - 9.171 SoC(k)^3 + 23.41 SoC(k)^4 - 31.96 SoC(k)^5 + 21.71 SoC(k)^6 - 5.78 SoC(k)^7, \quad (15)$$

$$C_{Th}(k) = 104.9 - 311.3 SoC(k) + 277.9 SoC(k)^2 - 2.435 SoC(k)^3 - 111.4 SoC(k)^4 + 44.86 SoC(k)^5 - 1.53 SoC(k)^6 + 0.108 SoC(k)^7. \quad (16)$$

A. Optimization Results

For the proposed method, when optimizing with different MSCC charging current sets and varying initial SOC conditions, the results obtained from the algorithm are illustrated in Fig. 8. As shown in Fig. 8, the final charged capacity consistently converges around 1800 mAh, which corresponds to approximately 90% of the battery capacity. This value lies within the target capacity range defined by the optimization algorithm.

TABLE II: Battery Specifications

Parameter	Value	Unit
Capacity	2000	mAh
Nominal Voltage	3.7	V
Charge Cut-off Voltage	4.2	V
Discharge Cut-off Voltage	2.7	V
Maximum Charge Current	1.0	C
Operating Temperature	-20 to 60	°C

TABLE III: Model Parameters

Parameter	a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7
U_{oc} (V)	2.896	9.563	-57.92	188.2	-340.7	347.5	-184.2	38.95
R_o (Ω)	0.039	-0.131	0.507	-1.129	1.679	-1.764	1.157	-0.331
R_p (Ω)	0.048	-0.318	2.075	-9.171	23.41	-31.96	21.71	-5.78
C_p (kF)	104.9	-311.3	277.9	-2.435	-111.4	44.86	-1.53	0.108

B. Simulation Results

To validate the accuracy of the proposed optimization algorithm, simulations are performed and the results are compared with conventional charging strategies. In particular, the charging time and charged capacity are evaluated between the MSCC method and the conventional CC-CV approach, where the CV stage is set at 4.2 V with a cut-off current of 0.1 A equivalent to 5 % of 1 C, and the CC stage is performed at 2 A. Furthermore, comparisons are made between the optimized MSCC at 0% SOC and the SOC-dependent optimized MSCC parameter sets.

As illustrated in Fig. 9 and Fig. 10, the proposed method consistently achieves shorter charging times than both the conventional CC-CV method and the traditional MSCC strategy with fixed parameters for charging from 0% to 90% SOC. Significant improvements are observed at 15% SOC, 45% SOC, and 80% SOC. Specifically, the proposed method reduces the charging time by up to 280 s compared to traditional MSCC at 45% SOC and by up to 320 s compared to the conventional CC-CV method. Meanwhile, the charged capacity differs only slightly, with less than 20 mAh compared to traditional MSCC and less than 30 mAh compared to the CC-CV method.

VI. CONCLUSIONS AND FUTURE WORKS

In this paper, the proposed MSCC method, which adapts to the measured SOC at the beginning of charging, demonstrates a significant reduction in charging time—up to nearly

Initial SOC (%)	I1 (A)	I2 (A)	I3 (A)	I4 (A)	I5 (A)	Time (s)	Capacity (mAh)
0	2	1.117	0.711	0.430	0.242	3662.09	1806.01
5	2	1.184	0.777	0.503	0.288	3419.42	1803.22
10	2	1.397	0.867	0.515	0.331	3190.56	1800.59
15	2	1.384	0.880	0.577	0.381	2962.41	1797.61
20	2	1.253	0.692	0.470	0.274	2899.00	1804.08
25	2	1.239	0.747	0.450	0.284	2702.37	1803.47
30	2	1.187	0.680	0.431	0.249	2570.03	1805.57
35	2	1.204	0.621	0.411	0.269	2368.21	1804.37
40	2	1.284	0.793	0.479	0.292	2152.58	1802.98
45	2	1.409	1.024	0.764	0.566	1759.32	1786.53
50	2	1.212	0.859	0.539	0.290	1800.45	1803.05
55	2	1.146	0.663	0.427	0.269	1642.22	1804.33
60	2	1.191	0.773	0.491	0.276	1453.80	1803.92
65	2	1.335	0.851	0.439	0.242	1328.12	1806.02
70	2	1.325	0.772	0.518	0.282	1089.68	1803.57
75	2	1.244	0.679	0.478	0.305	881.87	1802.16
80	2	1.353	1.084	0.675	0.418	598.53	1795.38
85	1.65	1.299	1.299	0.517	0.246	598.24	1805.75

Fig. 8: The optimization results base on different initial SOC

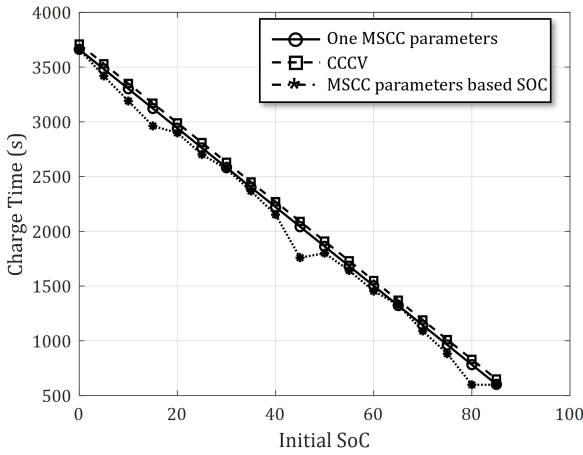


Fig. 9: Comparison charging time between charging methods.

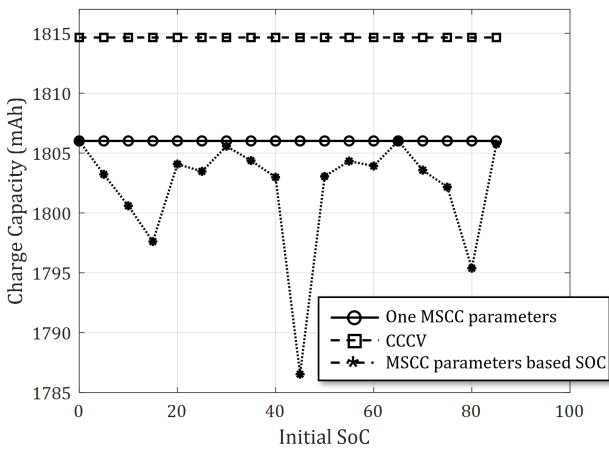


Fig. 10: Comparison charging capacity between charging methods.

320 s compared with the conventional CC-CV method and 280 s compared with the traditional MSCC optimization that only considers charging from 0% SOC. This improvement is achieved with only a minor trade-off in delivered capacity, i.e., approximately 30 mAh compared to CC-CV and 20 mAh compared to the conventional MSCC optimization. Furthermore, the results indicate that the initial charging current can be reduced while still maintaining optimized charging time. This feature has the potential to alleviate current-induced stress on the battery and thereby extend its cycle life.

Nevertheless, several limitations remain. In particular, the SOC estimation in this study is solely based on the relationship between SOC and OCV, without incorporating more advanced estimation techniques such as the Kalman Filter.

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